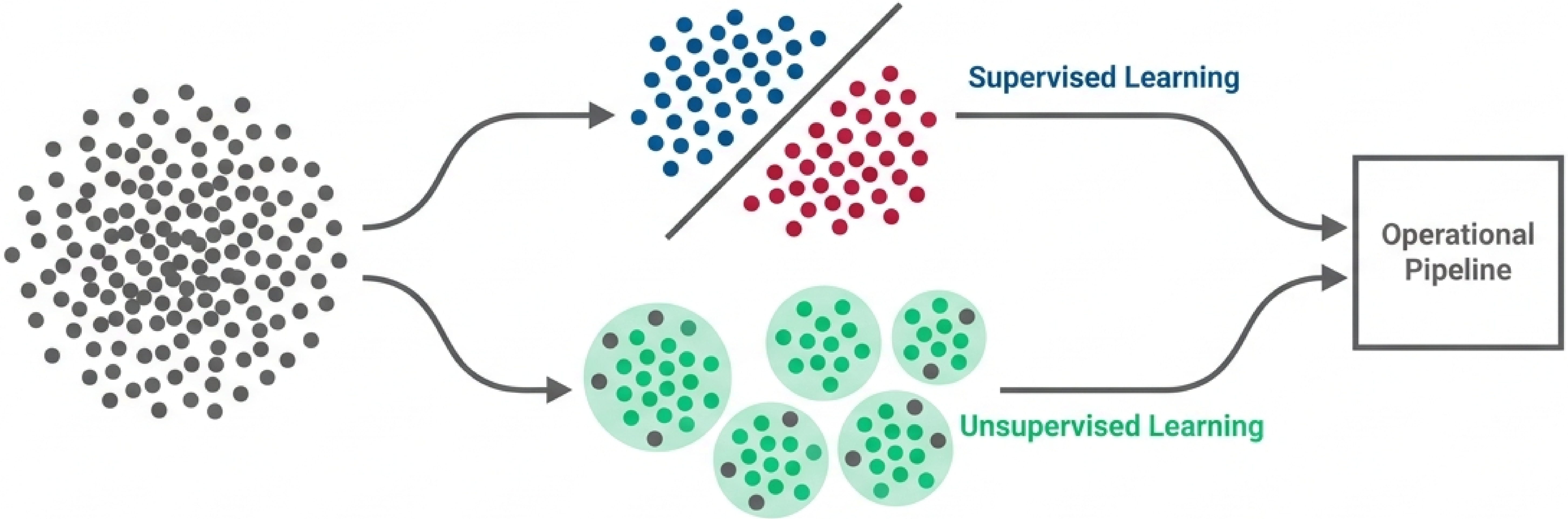
Introduction to Machine Learning Paradigms

Supervised Learning, Unsupervised Learning, and the Operational Pipeline

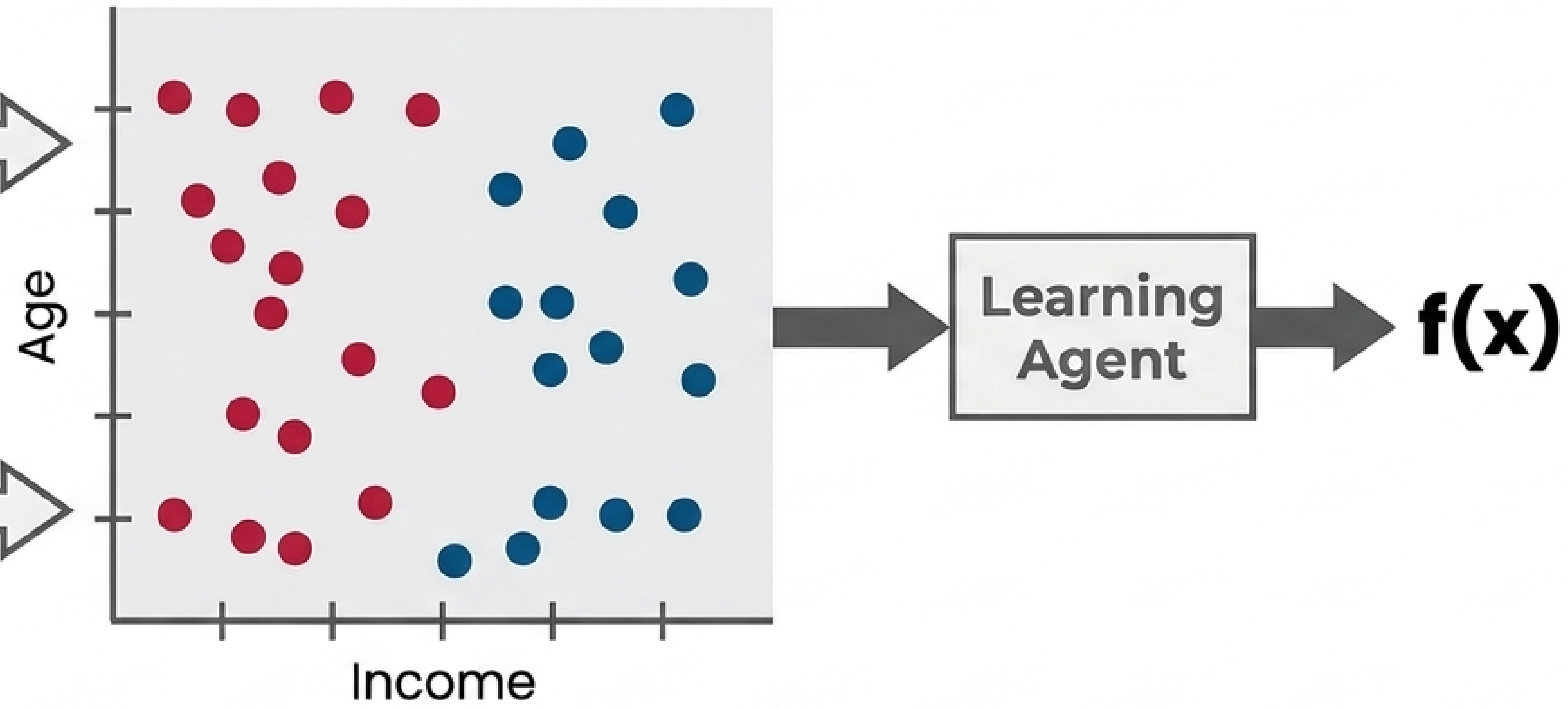


Supervised Learning (The Guided Path)	Unsupervised Learning (The Exploratory Path)
Input	**Input**
Data includes explicit answers (Labeled Data).	Data contains no provided answers (Unlabeled Data).
Mechanism	**Mechanism**
Learns from Experience + Known Outcomes.	Learns from Experience only.
Goal	**Goal**
Predict outcomes for new, unseen data.	Discover hidden structures, inherent groupings, or rules.
Core Tasks	**Core Tasks**
Classification & Regression.	Clustering & Association Rule Mining.

Supervised Learning: Deducing the Mapping Function

Feature Vectors (X): The descriptive attributes of the data (e.g., Age, Income).

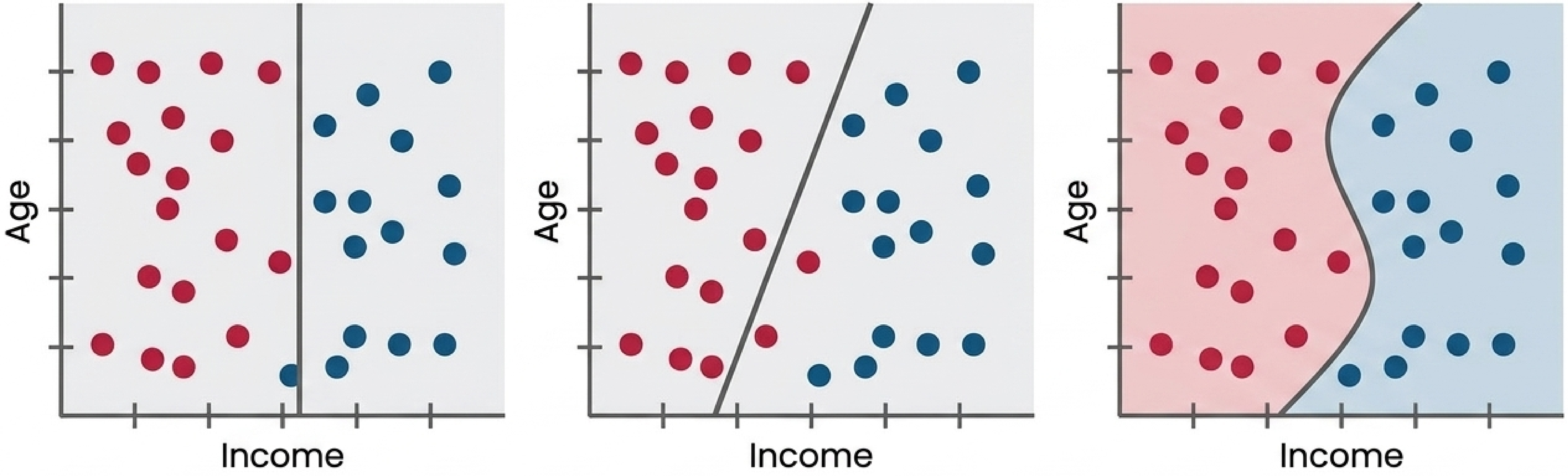
Target Labels (Y): The known, desired outcome (e.g., Red = 'Won't Buy', Blue = 'Will Buy').



The Objective: Deduce a mapping function f(x) that accurately maps inputs to the correct discrete or continuous output space.

Supervised Task 1: Classification

Predicting discrete, categorical outputs by drawing decision boundaries.

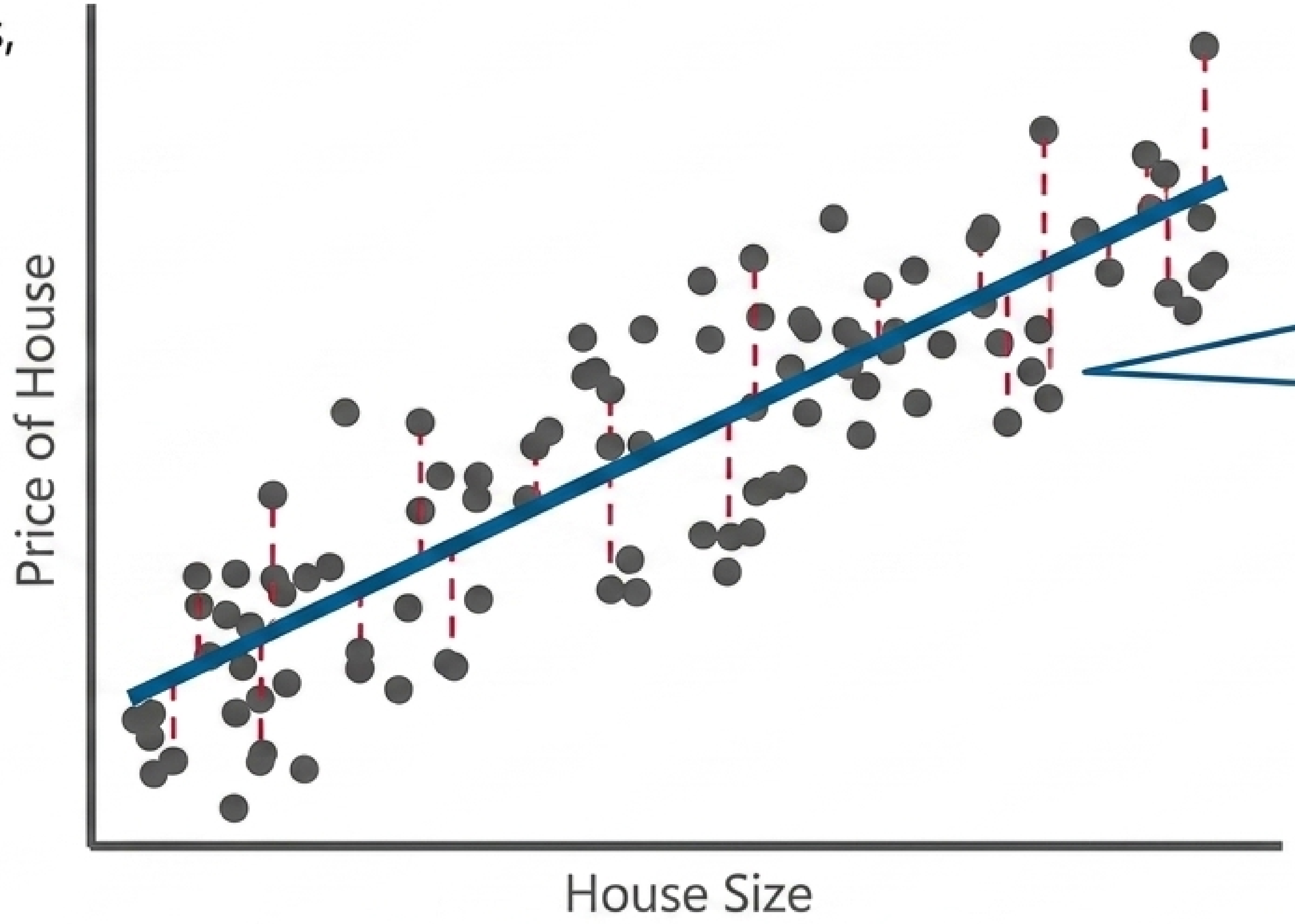


Real-world Applications:

Credit card fraud detection (Valid vs. Fraudulent)	Customer purchase prediction (Yes/No)	Medical diagnosis risk analysis (Disease/No Disease)

Supervised Task 2: Regression

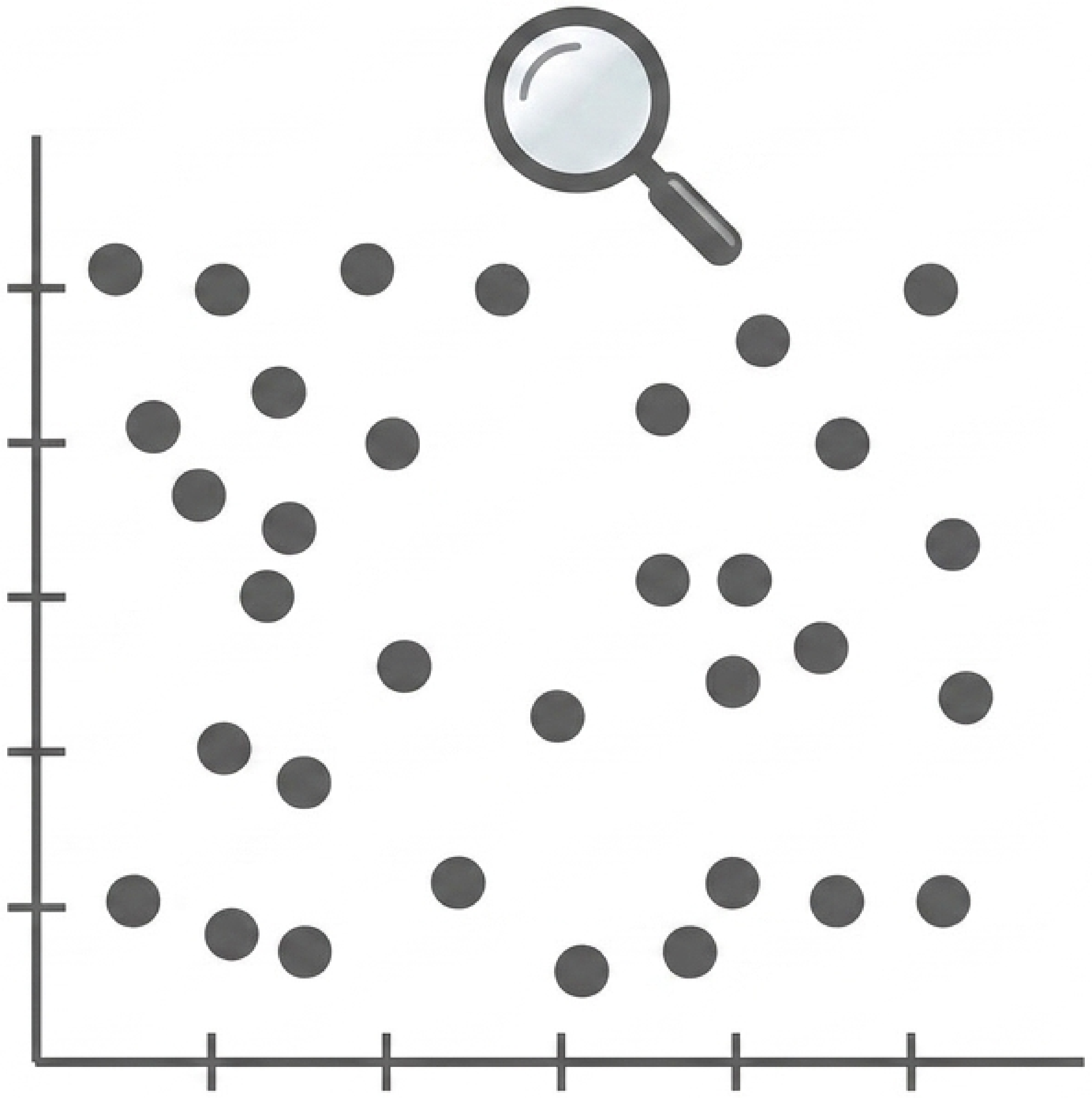
Predicting continuous, real-valued outputs.



The Optimization Goal: Minimize the sum of squared errors to find the best-fitting line or curve through the feature space.

Real-world Applications: Time-series temperature prediction | Real estate price forecasting | Trend analysis (linear vs. exponential scaling)

Unsupervised Learning: Letting the Data Speak



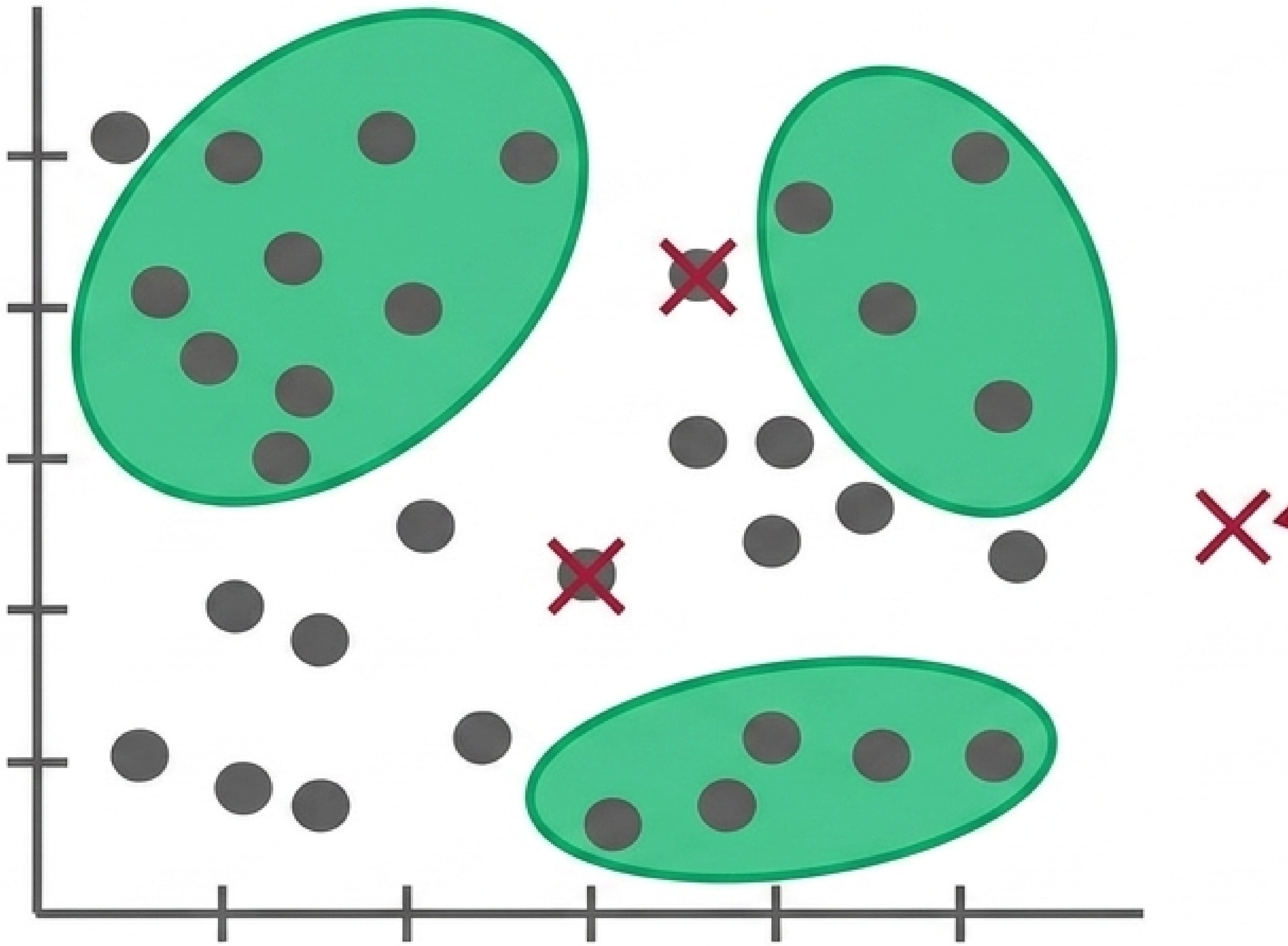
Input Components:
Feature Vectors (X) only.

The Paradigm Shift:
Without Target Labels (Y), the algorithm cannot be guided to a specific answer.

The Objective:
Discover inherent groupings, structural densities, or hidden association rules without human-provided categories.

Unsupervised Task 1: Clustering

Finding cohesive, coherent groups of similar data points within the input space based on an inherent bias (e.g., assuming groups form ellipsoid shapes).

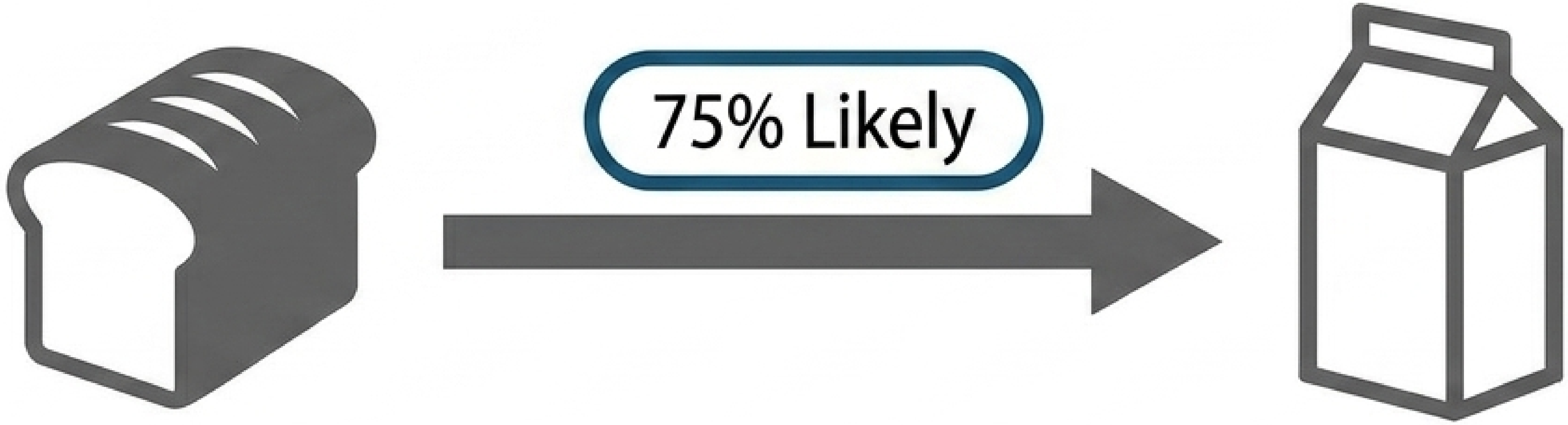


Key Insight: Clustering simultaneously identifies the norm (the groups) and the anomalies (the outliers).

Real-world Applications: Discovering distinct classes for customer segmentation | Image pixel region discovery (segmenting sky, sea, sand) | Anomaly detection

Unsupervised Task 2: Association Rule Mining

Discovering frequent patterns and conditional dependencies within datasets.



The Logic: If Event A happens -> Event B is highly likely to follow.

Real-world Applications:

Market basket analysis (predicting co-occurring purchases)	Time-series trigger events	Fault diagnostics

The Operational Pipeline: Data Preparation

Raw Data

Income	Age
Income: 80,000	Age: 25
Income: 150,000	Age: 42

Preprocessing Engine

Normalized Data

Income	Age
Income: 0.40	Age: 0.25
Income: 0.75	Age: 0.42

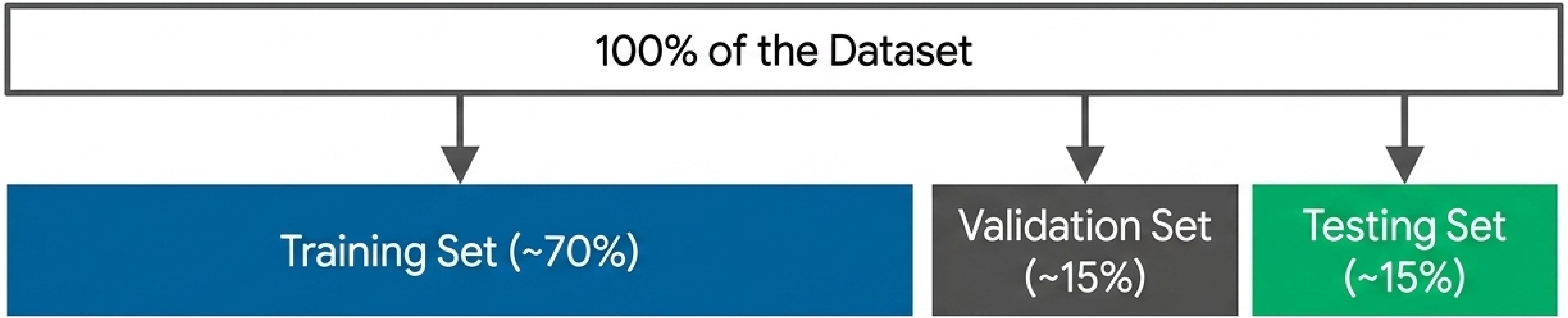
Data Cleaning:

Resolving missing values, identifying outliers, and handling inconsistencies.

Normalization:

Algorithms are mathematically naive. If Income is measured in thousands and Age in tens, the algorithm will disproportionately weigh Income due to magnitude. We scale all features into the same range (e.g., 0 to 1) to ensure numerical stability.

The Operational Pipeline: Data Splitting

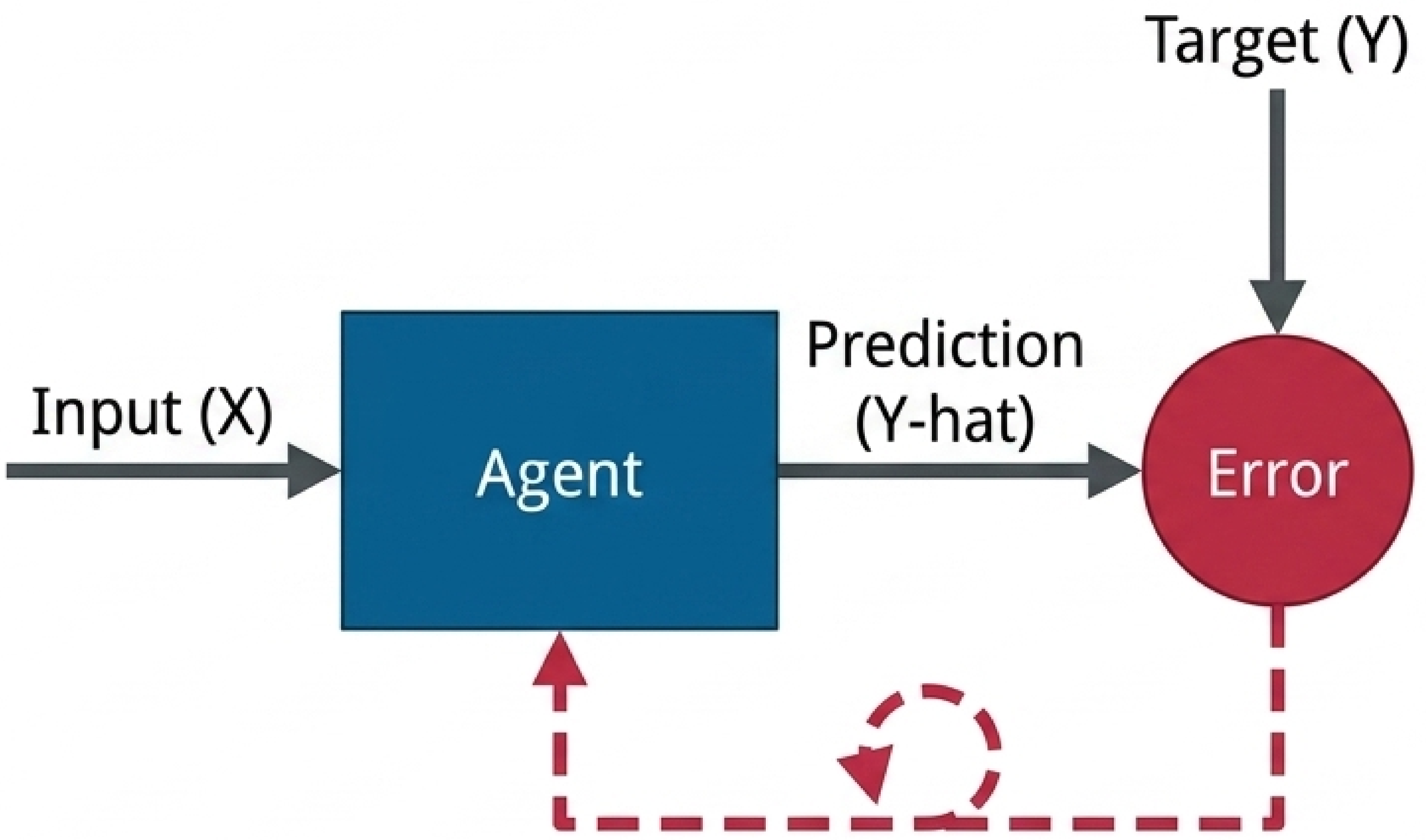


Fed into the algorithm to learn the mapping function. (Analogy: Doing the homework).

Used to evaluate the model during training, allowing for parameter refinement and tuning. (Analogy: Taking a practice test).

Unseen data held back until the very end to report the true, unbiased performance of the finalized model. (Analogy: The final exam).

The Iterative Learning Loop



Empirical Risk Minimization:

1. Predict: The agent makes an initial, often flawed, prediction (Y-hat).

2. Evaluate: Calculate the error against the true target (Y).

3. Adjust: Use the calculated error to iteratively update the agent's internal mathematical parameters.

This loop repeats until the error is minimized.

Evaluating Success: The Metrics of Machine Learning

Classification Metric: Accuracy

Question: Did we predict the correct discrete category?

Measure: Error Rate (Total misclassifications / Total instances).

Regression Metric: Mean Squared Error (MSE)

$$\text{Cost} = \sum (\text{Predicted} - \text{Actual})^2$$

Question: How far off were our continuous predictions?

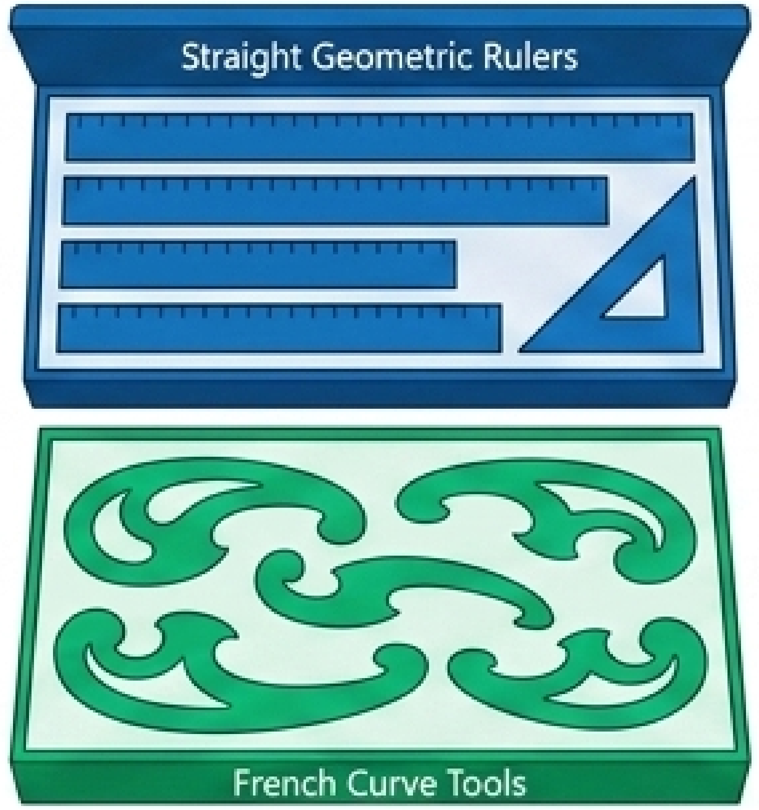
Measure: The sum of squared prediction errors.

Why square it?: Errors can be positive or negative; squaring ensures we penalize magnitude equally regardless of direction, while disproportionately punishing large outliers.

Core Principle: The Necessity of Inductive Bias

Insight: To predict unseen data, a model cannot simply memorize the training set; it must make inherent assumptions about the world. Bias is a feature, not a bug.

Language Bias



Constraining the type of functions the algorithm is allowed to draw (e.g., assuming the boundary must be a perfectly straight line).

Search Bias

$f1(x) = \ldots$

$f2(x) = \ldots$

$f3(x) = \ldots$

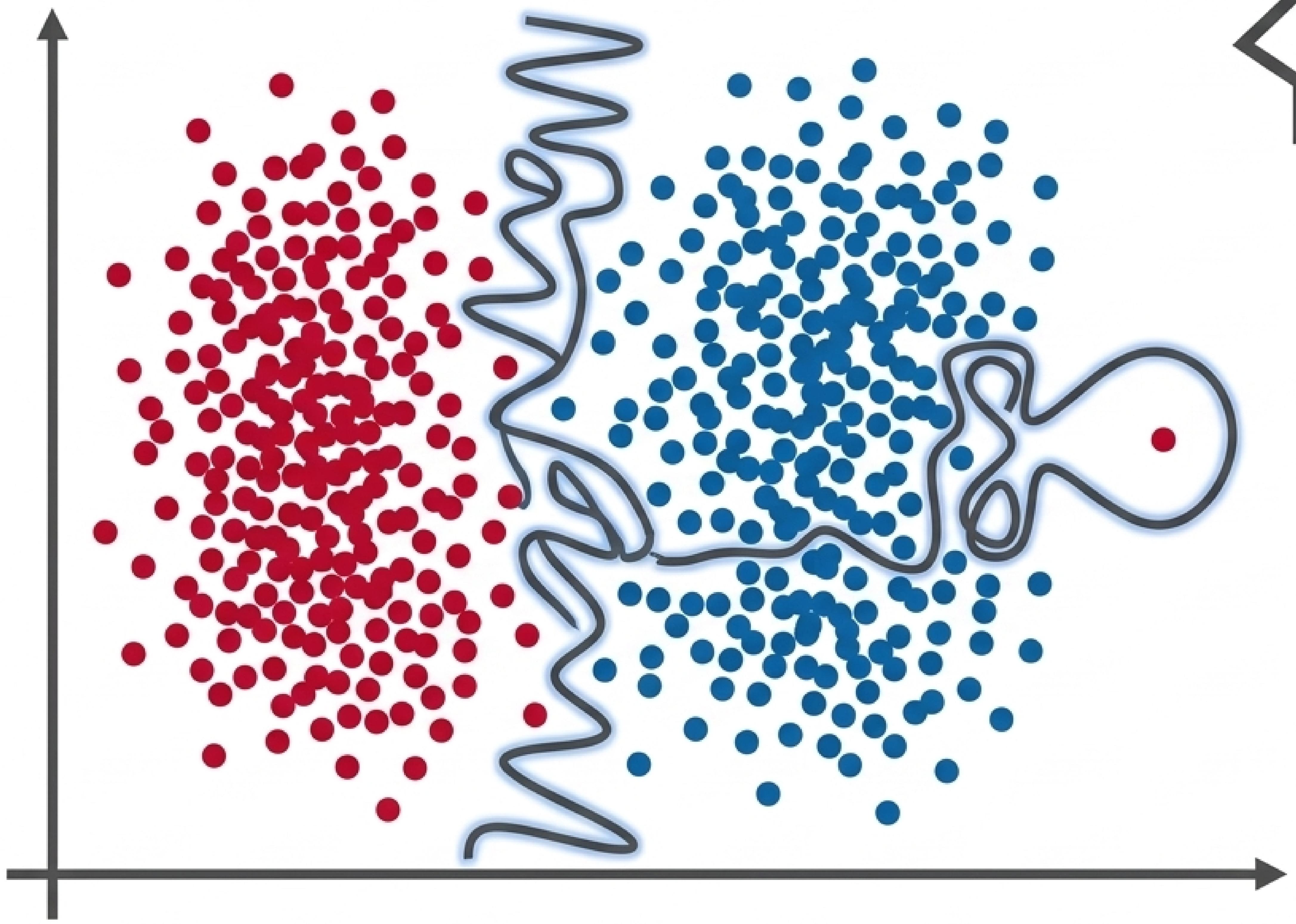
$f4(x) = \ldots$

$\ldots$

$fn(x) = \ldots$

Constraining the order in which the algorithm explores these possible functions to find the best fit efficiently.

The Ultimate Pitfall: Overfitting



Definition:

When a model is excessively complex, it stops learning the underlying pattern and starts memorizing the noise in the training data.

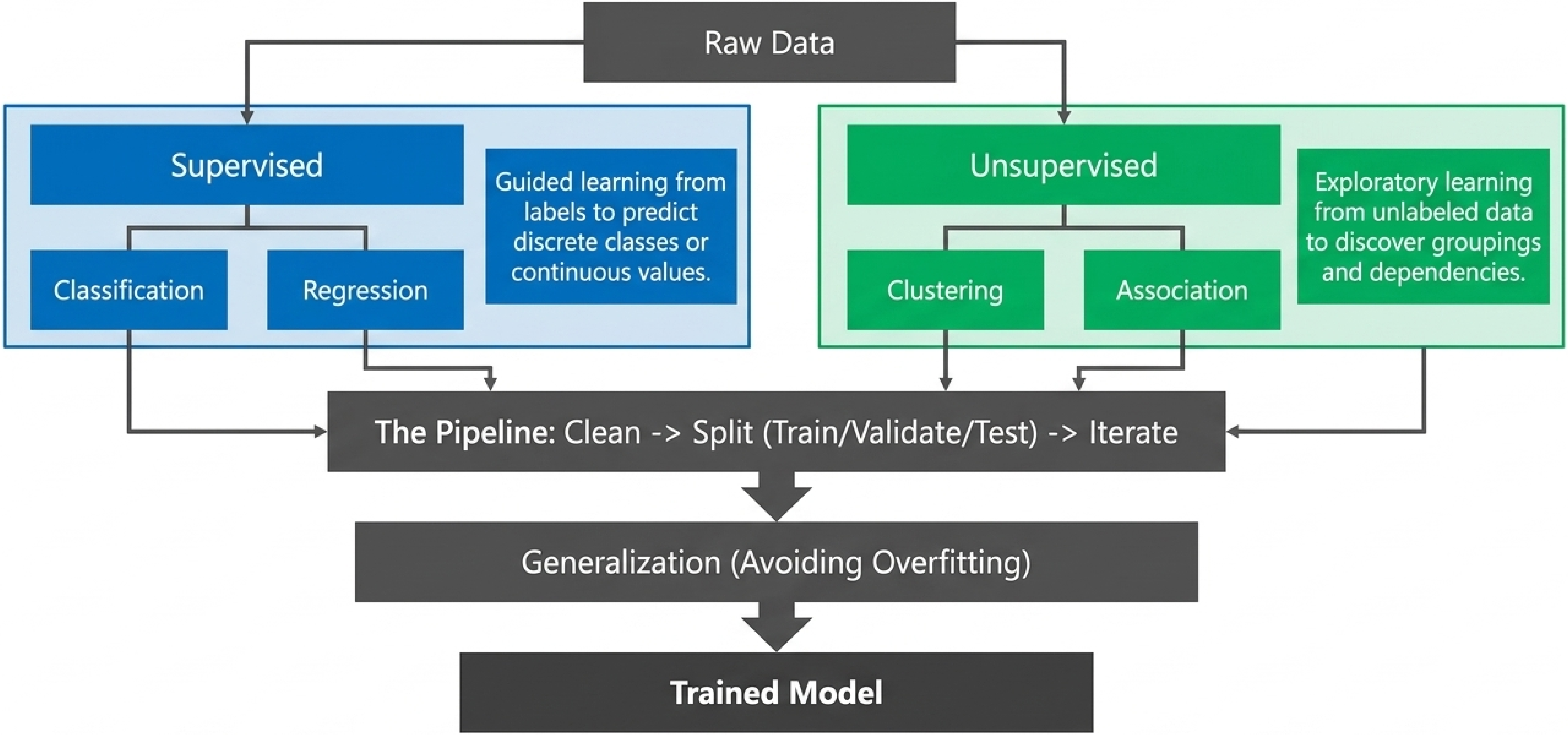
The Illusion of Perfection

The squiggly boundary line achieves 100% accuracy on the Training Set by memorizing anomalies, but it will fail catastrophically at generalizing to the unseen Validation or Test sets.

The Goal

Balance model complexity with data quality to capture the true trend.

The Architecture of Machine Learning



The Ultimate Objective: Leverage inductive bias to build models that generalize perfectly to the unseen future